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# Sizing Reserves within a Landscape: The Roles of Villagers' Reactions and the Ecological-Socioeconomic Setting

*Elizabeth J. Z. Robinson, Heidi J. Albers, and Jeffrey C. Williams*

**ABSTRACT.** *Traditionally, siting and sizing decisions for parks and reserves reflected ecological characteristics but typically failed to consider ecological costs created from displaced resource collection, welfare costs on nearby rural people, and enforcement costs. Using a spatial game-theoretic model that incorporates the interaction of socioeconomic and ecological settings, we show how incorporating more recent mandates that include rural welfare and surrounding landscapes can result in very different optimal sizing decisions. The model informs our discussion of recent forest management in Tanzania, reserve sizing and siting decisions, estimating reserve effectiveness, and determining patterns of avoided forest degradation in Reduced Emissions from Deforestation and Forest Degradation programs. (JEL Q23)*

## I. INTRODUCTION

Criteria over sizing and siting decisions for protected areas (PAs) such as forest reserves have evolved as policy makers have broadened their mandates to include nearby rural households' welfare and surrounding landscapes, and as economists and ecologists have recognized the costs PAs impose on nearby households in developing countries. These costs include the costs of purchasing land for a reserve system; the problems of spillovers and leakage as villagers displace their collection of nontimber forest products (NTFPs) into other forests; and the costs of enforcing the reserve mandate (Faith et al. 1996; Ando et al. 1998; White and Martin 2002; World Wildlife Fund 2002; Robinson and Albers 2006). In this paper, we explicitly look at how costs that PAs impose on nearby households

and forests in developing countries can be incorporated into PA sizing and siting decisions and into new initiatives such as Reduced Emissions from Deforestation and Forest Degradation (REDD) programs. We develop a model for the optimal size of a protected forest that recognizes that, within the context of the specific conservation mandate, the availability of markets, and the ecological setting, reserve policy imposes economic costs on rural people, and in turn their reactions to PA policy impose ecological costs.

Our paper joins a large and growing literature that addresses sizing and siting of PAs. The Single Large or Several Small literature and the Reserve Site Selection literature typically focus on how to create a PA so as to achieve an ecological objective such as "covering" a number of species. Although policies to address poverty and conservation simultaneously abound, sizing decisions for PAs rarely consider nearby people's welfare. Similarly, although so-called landscape approaches take into account the value of forests both within and outside PAs (Faith et al. 1996; White and Martin 2002; World Wildlife Fund 2002), the sizing decisions do not rely on a behavioral model that connects the PA's size to the ecological impact on buffer zones and unprotected forests.

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Economists have expanded these literatures by incorporating the cost of purchasing land for a reserve system, which changes the optimal pattern of conservation (Ando et al. 1998; Armsworth et al. 2006). But most new PAs sit in developing countries (IUCN 2008), where other costs are more relevant (Naidoo and Adamowicz 2006; Naidoo et al. 2006). Nor does the literature consider how those costs in developing countries inform the optimal size, focusing rather on siting decisions. In developing countries, people typically extract fuelwood and food from forests for subsistence use or for sale, and these resources can contribute significantly to the effective income of these communities (Cavendish 2000). Establishing PAs, or reestablishing access restrictions in previously established but neglected PAs, places a burden on these local communities, who must change their behavior in response (Lokina and Robinson 2008; Ferraro 2002). They may decrease the amount of the resource they use overall, replace some of their extraction with purchases from a market or by growing their own, or displace their extraction activities to portions of the forest outside of the PA—a classic spillover effect. These decisions, which have both welfare effects and ecological effects in neighboring forests, depend on whether the resource has some minimum requirement, on the opportunity cost of labor, and on the ease of market access, which, in turn, affect the welfare impact of the PA on nearby rural people.<sup>1</sup>

Lokina and Robinson (2008) find deterioration in unprotected forests following the reenforcement of Tanzania's restrictions on access to some protected forests. Ferraro (2002) describes an "intensification" of resource use in a peripheral zone outside of a park in Madagascar. Depending on the type of ecosystem, "concentrating previously dispersed . . . activities into certain parts of the forest may actually increase the negative ecological impact" (Lewis 2002, 9).

Our paper and model are particularly motivated by our own experiences of PAs in a range of developing countries, most recently

in Tanzania, where PAs are being reestablished through participatory forest management (MNRT 1998, 2002a, 2002b; Kajembe, Nduwamungu, and Luoga 2005; FBD 2006). For example, Tanzania's Amani Nature Reserve, which fits the criteria for an IUCN PA Category II (Burgess and Rodgers 2004), must be protected from any extraction anywhere, according to national regulations. However, when the Amani management committee recognized that villagers did not have alternative forests from which to collect NTFPs, nor many trees on private land, they effectively reduced the size of the PA by permitting extraction, particularly fuelwood, which is collected by almost all households, from a *de facto* buffer zone inside the perimeter of the reserve. We use our model to explore the basis of the apparent conflict between national regulations for strict conservation and park managers' implementation of either smaller parks or less-restrictive parks.

In the model that follows, we focus on three key elements of sizing and siting decisions: the forest department's mandate, variation in ecological settings, and variation in the socioeconomic setting. We consider four stylized forest department (FD) mandates, which reflect the range found in developing countries. Our first, the PA-only mandate, mimics "old-school" conservation by considering only the ecological benefits generated within the PA itself. Our second, a landscape mandate, takes the newer and broader perspective in conservation biology in which the FD considers values derived from both the PA and unprotected areas (Faith et al. 1996; White and Martin 2002). In distinguishing these two mandates, we are not taking sides in the debates, such as Locke and Dearden (2005) versus Martino (2005), about which mandate better protects wild biodiversity, but rather we investigate what those two stylized mandates imply about the optimal size of the PA. To incorporate calls for environmental planners to consider rural livelihoods (such as by the Durban Accord [2005]), we consider a third FD with a PA-local welfare mandate, and a fourth FD with a landscape-local welfare mandate.

We account for variations in ecological setting by developing a set of four ecological

<sup>1</sup> The same point can be made about those cases, studied by Milne and Niessen (2009), in which villagers are paid to respect a PA.

damage functions (EDFs), which depict a range of ecological reactions to degrading activities in the unprotected zone. For a “pristine-only” forest, ecosystem services are zero unless the forest is pristine. This extreme EDF could proxy for a particularly fragile ecosystem or a forest that protects a highly endangered species. A “biomass-proportional” forest—perhaps one where carbon sequestration is important—provides ecosystem services proportionate to the sheer quantity of biomass. An “ecoservices” forest—perhaps offering biodiversity provides less value if it is degraded but does not lose its value entirely. Finally, an “ecothreshold” forest values biomass according to a logistic function, such that a small amount of extraction has little impact on the ecosystem services provided by the forest but below some critical level of degradation the forest loses most of its value. Many ecosystem services display this kind of threshold. For example, hydrological benefits appear insensitive to slight degradation but beyond some point the degraded forest contributes little to water flow control (Wu and Boggess 1999). Similarly, many species thrive in mildly degraded forests but become locally extinct below some critical degradation of habitat (Smith, Ahern, and McDougal 1998).

Following Robinson, Williams, and Albers (2002), a representative villager makes a labor allocation decision between resource extraction in the unprotected zone and other activities, based on the socioeconomic setting. The villager’s reaction to the size of the reserve is a function of her resource requirement, market access costs, resource abundance, a resource extraction production function, and the opportunities for nonextraction activities. Parameters that depict the cost of accessing NTFP markets and the opportunity cost of labor allow us to consider a range of market settings.

We incorporate three costs in determining the optimal size of a PA: the welfare losses for rural people; the displacement-related ecological costs in an unprotected zone; and enforcement costs. With this third cost we recognize that FDs may not be able to protect as large an area of forest as they consider optimal. We solve an optimal PA sizing model that recognizes in its leader-follower game

structure that villagers react to the PA. We explore the decision for four policy mandates, four ecological settings, and a range of market parameters.

## II. MODEL

In our model, the FD wants to establish, or reestablish, a PA according to a particular forest mandate. Villagers respond to a PA of a particular size by choosing how intensively to harvest NTFPs in the remaining unprotected zone, which determines the level of degradation there.<sup>2</sup> The ecological cost of this extraction, or degradation, in the unprotected zone is reflected in the EDF, which values the per-unit-width ecological contribution of the forest as a function of the level of degradation. In sum, we have a strategic interaction between the FD and the villager that we model as a Stackelberg interaction.

### The FD’s Optimization

The FD chooses how large a PA to enforce within a finite area of forest and, therefore, how large a zone to create from which villagers can collect NTFPs. Without loss of generality we assume the forested area to be one dimensional, and so we consider a forest of total width  $\bar{X}$ , which the FD allocates between pristine forest of width  $X_P$ , the PA, and an unprotected zone of width  $X_B$  (such that  $X_P + X_B = \bar{X}$ ). The ecological contribution of each of the zones is the product of the width of the zone multiplied by per-unit-width ecological contribution,  $E$ , which is a function of the resource density in the particular area and the specific EDF.<sup>3</sup> We set the resource density

<sup>2</sup> In reality, villagers may continue to extract from the PA illegally (Robinson and Albers 2006). In this paper, for ease of exposition, we assume that the extraction ban is fully enforced, thereby implying more intense harvesting in the buffer zone and higher ecological benefits from the PA than would occur with incomplete enforcement. Albers (2010) models patterns of enforcement (incomplete and complete) and illegal extraction, and their interaction with enforcement budget constraints.

<sup>3</sup> The model is also unidimensional in the sense that many species are compressed into one index. As noted by Oluput, Barigira, and McNeilage (2009) in their study of Bwindi Impenetrable National Park, Uganda, the effect on an individual species depends on its inherent rarity, the

in the pristine PA as  $m$  (such that the total quantity of resource is  $mX_P$ ) and let  $E(m) = 1$  for all EDFs, so that the ecological contribution of the PA is always  $X_P$ . The resource density in the unprotected zone is  $m - h(X_B)$ , where  $h(X_B)$  is a representative villager's harvest intensity, derived from her reaction function (below), and so the unprotected zone's ecological contribution is  $X_B E(m - h(X_B))$ . For enforcement costs, we assume a simple linear function,  $F(X_P) = eX_P$ , where  $e$  is sufficiently small to merit protecting some area.<sup>4</sup>  $P(X_P)$ , derived from a representative villager's reaction function, is the implicit penalty, or welfare cost, imposed on villagers due to the introduction of the PA into an area of forest to which they previously had access to collect NTFPs. The FD therefore chooses  $X_P$ , the width of the PA, according to

$$\max_{X_P} \{W\} = \max_{X_P} \{\delta_1 X_P + \delta_2 (\bar{X} - X_P) E(m - h(X_P)) - \delta_3 P(X_P) - \delta_4 e X_P\}, \quad [1]$$

where the deltas reflect the components of the FD's mandate as described above:  $\delta_1$  and  $\delta_4 = 1$  and  $\delta_2$  and  $\delta_3 = 0$  for the PA-only mandate;  $\delta_1$ ,  $\delta_2$ , and  $\delta_4 = 1$  and  $\delta_3 = 0$  for the landscape mandate;  $\delta_1$ ,  $\delta_3$ , and  $\delta_4 = 1$  and  $\delta_2 = 0$  for the PA-welfare mandate; and  $\delta_1$ ,  $\delta_2$ ,  $\delta_3$ , and  $\delta_4 = 1$  for the landscape-welfare mandate.

The first term on the right-hand side of equation [1] is the ecological contribution of the PA, the second is the ecological contri-

bution of the unprotected zone, the third is the welfare penalty imposed on the villagers, and the fourth is the enforcement cost of protecting the PA from NTFP collection.<sup>5</sup> The ecological contribution of the unprotected zone depends on the EDF. For the particular ecological settings described above,

$$E(m - h) = \begin{cases} 1 & \text{if } h = 0 \text{ else } = 0 & \text{for the "pristine-only" EDF} \\ (m - h)/m & & \text{for the "biomass-proportional" EDF} \\ [(m - h)/m]^\beta, \beta < 1 & & \text{for the "ecoservices" EDF} \\ 1/\{1 + \theta \exp[-b(m - h)/m]\} & & \text{for the "ecothreshold" EDF.} \end{cases} \quad [2]$$

Differentiating equation [1] with respect to  $X_P$ , and recognizing that  $\delta_1$  and  $\delta_4 = 1$  for all of the four mandates, we can write

$$W' = 1 - \delta_2 \left[ (\bar{X} - X_P) \frac{\partial E}{\partial h} \frac{\partial h^*}{\partial X_P} + E(m - h) \right] - \delta_3 P'(X_P) - e. \quad [3]$$

Equation [3] demonstrates explicitly what may seem obvious but is rarely acknowledged in the literature: the reaction of local people to the introduction, or reintroduction, of a PA is a key determinant of the impact and, therefore, the success or failure of a PA policy. The response of local people, their harvest intensity  $h$ , comes into both the second and third terms on the right-hand side of equation [3], and therefore, if either  $\delta_2$  or  $\delta_3 = 1$ , the reaction of local people matters. The ecological impact of the policy within the unprotected zone, the second term, depends on both the forest-specific EDF and how the PA changes the intensity of villagers' harvesting. The third term accounts for the direct impact of the PA on villagers' welfare.

changes in topography, and whether it prefers "disturbed" landscapes. The model attempts, nonetheless, to clarify their conclusion, namely: "Including edge-related distribution in assessments of multiple-use zones may be informative in defining interior bounds for multiple-use zones. This is important because a zone that is too wide or too deep is only likely to encourage illegal activity without achieving much of the intended purpose" (p. 1144).

<sup>4</sup> Currently little research informs the shape of enforcement costs as a function of the area of a reserve. Robinson and Albers (2006) show that the costs of enforcement can disproportionately increase with increasing size of the PA, because a larger PA implies that villagers have less area from which to collect NTFPs and so put more pressure on the forest boundary. However, enforcement economies of scale, such as shared ranger stations and vehicles, could lead to the opposite result. (See Ferraro and Kramer 1997; Hallwood 2005; Robinson and Albers 2006; Robinson, Mahapatra, and Albers 2010; Albers 2010).

<sup>5</sup> In this paper, the unprotected zone is simply the area of forest outside of the PA and is under no active management. There are no enforcement costs here, and so the zone is de facto open access. Because we are considering only one period, we do not explore the implications for each zone of ongoing degradation in the unprotected forest. In other work, we explicitly model the degree of effectiveness of community resource management in forested areas and how degradation matters over time (Robinson, Albers, and Williams 2008). In Amani the local management refer to the unprotected area as a buffer zone.



We could simply introduce some reduced-form equation for  $\partial h/\partial X_p$ . However, our observations and other papers suggest that the reaction of local people depends on their access to NTFP markets and to labor opportunities (Robinson, Williams, and Albers 2002; Muller and Albers 2004).<sup>6</sup> Further, papers by Omamo (1998), Key, Sadoulet, and de Janvry (2000), Robinson, Williams, and Albers (2002), and Robinson, Albers, and Williams (2008) suggest that NTFP markets are far from perfect and are best modeled with the inclusion of some discrete fixed cost to market access. This cost introduces an asymmetry between the collection decision and the market decision and accommodates situations in which, over a range of parameters, villagers sell NTFPs to the market, buy from the market, or consume all they collect. As we demonstrate in the model, this asymmetry contributes to significant nonlinearities in the relationship between the width of the PA and the returns to particular FD mandates.

In the Appendix we motivate and develop the model of a representative villager's reaction function, following Robinson, Williams, and Albers (2002). We summarize the model here. The villager's key choice variable is  $w$ , the time spent extracting per unit distance, which translates into the harvest intensity  $h(w, m, \alpha)$  where  $\alpha$  is a parameter that takes into account extraction effectiveness. The villager's "type," whether she buys NTFPs ("supplementing"), sells NTFPs ("selling"), or collects her exact consumption requirement  $R$  ("subsistence"), is endogenous to the model.<sup>7</sup> The total harvest is  $H = hX_B$ , and the total time spent in the forest is  $(w + v)X_B$  at a time cost  $C = k[(w + v)X_B]^\gamma$ , where  $v$  is the time it takes the villager to walk a unit distance through the unprotected zone when

not extracting.  $k$  is a simple scaling parameter.  $y$  reflects the labor market conditions: the villager's labor can exhibit a constant ( $\gamma = 1$ ) or increasing ( $\gamma > 1$ , for an incomplete labor market) marginal opportunity cost of time.  $S$  is the quantity of NTFP sold by the household to the market,  $D$  the quantity purchased,  $p$  the price, and  $t$  transportation or market access costs. The villager contemplates a single-period optimization of net returns to her labor  $V$  subject to the resource requirement, such that  $V = (p - t)S - (p + t)D - C$ , which translates directly into the villager's welfare. Solving the model, we generate conditions for the three possible types that the villager could be: For a subsistence type,

$$h = \frac{R}{\bar{X} - X_p} \text{ and } w = \frac{R}{\alpha m [\bar{X} - X_p - R]}. \quad [4]$$

For a supplementing type,  $w$  is the solution to the first-order condition:

$$(p + t)\alpha m^2 = k\gamma[(w + v)(\bar{X} - X_p)]^{\gamma-1}(1 + \alpha mw)^2. \quad [5]$$

For a selling type,  $w$  is the solution to the first-order condition:

$$(p - t)\alpha m^2 = k\gamma[(w + v)(\bar{X} - X_p)]^{\gamma-1}(1 + \alpha mw)^2. \quad [6]$$

In general, there will be some range of PA widths for which the villager does not interact with the market, in which case the villager collects exactly her requirement and the harvest intensity is simply the resource requirement divided by the width of the unprotected zone (equation [4]). If the villager interacts with the market, her choice of  $w$  and, therefore, her harvest intensity depend also on  $t$ , the market-access costs, and  $\gamma$ , the labor market conditions (equations [5] and [6]).

### III. RESULTS AND LINKS TO TANZANIAN PA DECISIONS

In this section we present results from our model, emphasizing how mandates matter, how markets matter, and how the ecological consequences of degradation matter when making optimal sizing/siting decisions. The

<sup>6</sup> The opportunity cost of labor is also affected by regional circumstances. According to the study by Fu et al. (2004), of those who lived within the Wolong Biosphere Reserve (southwestern China) but had to compete for low-skilled construction jobs in the region, "most of the local excess labor went to the natural environment to collect resources for household use or sale to supplement the household income and kill time" (p. 795).

<sup>7</sup> A fourth "nonextracting" type, who purchase their full requirement from the market, is a subset of villagers who buy NTFPs from the market.

first-order conditions for the FD's optimal sizing decision, which we derived in equation [3], depend on the particular mandate and on the response functions for the villagers, which in turn depend on their endogenous type. Even keeping the model simple, there are many permutations because we have four FD mandates, four EDFs, and market assumptions for both NTFPs and labor. Few of these permutations result in neat analytical findings. Therefore we present here general results for each mandate followed by numerically solved results for various market and ecological settings while using the framework as a lens through which to view PA decisions in Tanzania and elsewhere. Throughout, we emphasize the key point of this paper: villager reactions to a PA, and the impact of degradation on the ecological contribution of the total forest area and the market conditions interact and play a large role and should be included in determining optimal PA sizing and siting decisions. Yet, they rarely are.

Tanzania's Amani Nature Reserve illustrates some permutations of the relationships in our framework.<sup>8</sup> First, a local market for the principal NTFP, fuelwood, does not exist.<sup>9</sup> Nearby villagers mainly extract fuelwood from Amani, and there are few alternatives, implying large market access costs,  $t$ .<sup>10</sup> Second, the local labor market is best characterized as incomplete, as elsewhere in Tanzania, implying  $\gamma > 1$  (Lanjouw, Quizon, and Sparrow 2001). Third, given that Amani is a biodiversity hot spot and important for ecosystem provisioning including watershed protection, the EDF would seemingly be the "ecoservices" or "ecothreshold" type. Fourth, Amani is designated a strict reserve, which implies a PA-only mandate. However, Amani's managers follow a PA-welfare mandate. One outcome of our model is that all of these

characteristics of the socioeconomic and ecological setting should contribute to the reserve sizing decision, but we have seen little evidence that Tanzanian, nor environment departments in most other countries, nor most academic researchers, incorporate this full range of information into their siting and sizing decisions.

### FD Mandates

From equation [3], with any EDF and with any reaction from rural people, an FD with the PA-only mandate ( $\delta_2$  and  $\delta_3 = 0$ ) always sizes the PA as large as the budget permits. Therefore, with an unlimited enforcement budget, the PA-only FD protects the full width of the forest. This statement is intuitive: the FD does not consider the impact on nearby villagers, nor does it attach any value to degraded land. These conditions fit with the description of many national parks throughout the world and the official PA-only mandate for Amani Game Reserve in Tanzania, in keeping with the land's official classification.

If we consider the Amani local manager's de facto PA-welfare mandate,  $\delta_2 = 0$ , and so again the EDF does not influence the optimal sizing decision. But the villager response matters for the welfare component:  $W' = (1 - e) - P'(X_P)$  and  $W'' = -P''(X_P)$ , where  $P'(X_P) \geq 0$ . As predicted by our model, in Amani the forest manager's informal adoption of a PA-welfare mandate resulted in a reduction in the size of the protected forest and the establishment of a de facto buffer zone—an area of unprotected forest where extraction is permitted—which was based on his local knowledge of the nearby landscape, specifically that there are no nearby forests from which the villagers can collect, nor markets from which the villagers can purchase fuelwood. Similarly, park managers in Khao Yai National Park, Thailand (KYNP), also deviate from national regulations. They have in effect reduced the size of the PA by allowing extraction for home use by nearby villages, thereby also demonstrating a PA-welfare mandate (Albers and Grinspoon 1997).

For both the official Amani PA-only mandate and the de facto PA-welfare mandate of the local manager, leakage is not a concern,

<sup>8</sup> Neither we nor the Tanzanian authorities have undertaken an analysis of the area to determine the exact parameters of our model. However, what is key is the characterization of the key parameters and identification of the EDF.

<sup>9</sup> Alternative forms of fuel have not been considered, such as the solar panels installed in households near the El Imposible National Park in El Salvador when access to the park became stricter (Balint 2006).

<sup>10</sup> In contrast, illegal logging is a major problem in the Tanzanian forests studied by Persha and Blomley (2009).

and so neither considers the EDF to determine its optimal strategy. However, as more attention is given to “landscape” approaches to forest management, such as if Amani were to be included in any REDD considerations, the official mandate could well be required to take into account the full landscape of protected and unprotected forested areas. In such a case, looking again at equation [3], for both the landscape and landscape-welfare mandates,  $\delta_2 = 1$  and we must take account of the impact of leakage on the contribution of the unprotected forest to the overall ecosystem:  $\partial E / \partial h^* \leq 0$ , and  $E(m - h) \geq 0$ . If villagers respond to a larger PA by harvesting more intensively in the unprotected zone (which they will do, so long as  $\partial h^* / \partial X_P > 0$ , implying a displacement effect), then the FD’s optimal strategy with either of these two mandates now depends on the EDF. Equations [3] through [6] together show how the NTFP and labor markets’ function influences the size of the displacement effect (that is, leakage/spillovers) and, therefore, the optimal PA size. But looking at these equations alone does not provide sufficient insight into the interactions between mandate, EDF, and markets. We next consider in more detail these interactions, bringing in additional practical examples of reserve management from Tanzania, Thailand, and China.

### Labor and NTFP Market Conditions

We first highlight three findings from the model with respect to labor and market conditions. First, if a villager neither buys nor sells NTFPs (sufficiently large  $t$ ), then for all  $\gamma$  (opportunity cost of labor), there is always a displacement effect:  $h'(X_P) > 0$ . Second, if the villager buys or sells NTFPs (sufficiently small  $t$ ) and  $\gamma > 1$ , there is also a displacement effect, albeit smaller than under autarkic conditions. Third, if the villager buys or sells NTFPs (sufficiently small  $t$ ) and  $\gamma = 1$  (perfect labor markets), there is no displacement effect ( $w = (1/\alpha)\sqrt{(p-t)\alpha/k} - 1/\alpha m$  and so  $h'(X_P) = 0$ ).<sup>11</sup> We illustrate this displacement

effect schematically in Figure 1 for zero, small, and large NTFP market access costs ( $t$ ) and for constant and increasing labor opportunity costs ( $\gamma = 1$  and  $\gamma > 1$ , respectively). Figure 1 confirms that the only conditions under which there is no displacement for all PA widths occurs when  $\gamma = 1$  and  $t = 0$ , that is, with complete and costless labor and NTFP markets. The varying slopes graphed in Figure 1 reveal the ambiguity in the second derivative of the villager penalty and extraction intensity with respect to the width of the PA, caused by the presence of NTFP market access costs.

For reserves such as Amani (large  $t$ ,  $\gamma > 1$ , corresponding to Figure 1, right-hand panels), the displacement effect is likely to be large, and for all but the smallest PAs, the penalty on nearby villagers high. Moreover, for our model calibration, efforts to improve the functioning of NTFP markets will reduce both the villager penalty and the displacement effect—that is, leakages—and be more effective than efforts to improve labor markets alone. In Amani, the FD has encouraged butterfly farming to give nearby residents alternative income. Elsewhere in Tanzania, beekeeping has been encouraged to compensate villagers for the loss of access to extractable forest resources. Alternative income projects improve welfare and can move some labor to the project activity and away from the extraction activity, but without local resource markets or tree planting initiatives that have had little success in Amani, villagers will continue to extract from reserves. In the case of KYNP, some neighboring villages have good labor prospects in nearby dairy facilities, yet others, on the other side of the park, are remote. According to our model, that heterogeneity should influence the size of the area where extraction is tolerated. Although the fact that markets matter for the interaction of rural people and PAs has been noticed before, our framework demonstrates that the interaction between labor and NTFP markets and optimal PA size is not a simple one and that the PA sizing decisions require at least as much in-

<sup>11</sup> Although we have chosen specific functional forms for the villager optimization, these findings are relatively

robust to a large number of different formulations of the villager optimization.



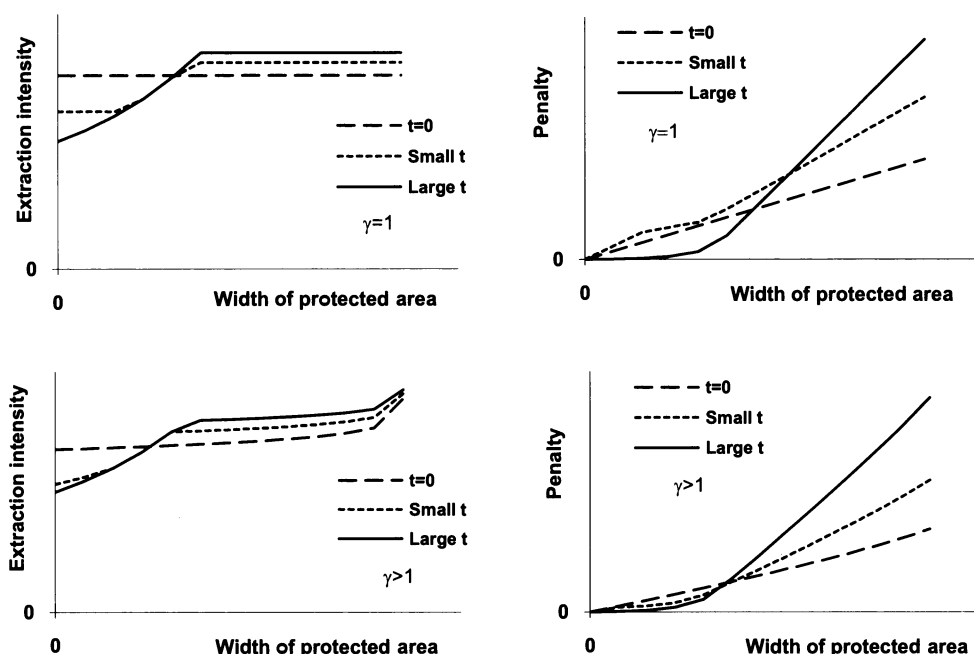


FIGURE 1  
Displacement Effects and Villager Penalty

formation about the local socioeconomic setting as about the ecological setting.

### Degradation, EDFs, and Markets

Even a FD with a landscape mandate, which does not account for the impact of the PA on villager welfare, must typically take into account the reaction of villagers, because how villagers react to the PA alters the environmental services within the unprotected forest. Figure 2 depicts the ecological contribution of the entire landscape as a function of the PA width for all four EDFs. That contribution varies with the market setting, and so we illustrate two extremes for markets: poorly functioning markets (left panel,  $\gamma = 1.3$ ,  $t = 1.2$ ) and well-functioning markets (right panel,  $\gamma = 1.0$ ,  $t = 0$ ). Figure 2 demonstrates a complex interaction between the ecological and socioeconomic setting in which PAs are sited. Where markets are functioning poorly, as is the case for Amani, for some intermediate widths of a PA and particular EDFs (ecoservices and ecothreshold), the ecological

contribution of the combined PA and unprotected zone actually decreases as the width of the PA is increased, providing theoretical support for Lewis's (2002) findings of ecologically costly concentration of degrading activities.

### Optimal Sizing Decisions

Having examined the mandate, market situation, and ecological setting, we now consider how these factors interact, possibly with a budget constraint, to determine the optimal reserve size. The ecothreshold EDF results in particularly interesting output and is relevant to Amani Reserve, and so we use it to illustrate in Figure 3 the optimal PA size under the two market settings of poorly functioning and well-functioning markets, as for Figure 2.<sup>12</sup>

<sup>12</sup> Both the ecoservices and ecothreshold EDFs result in particularly interesting and relevant output for forests such as Amani Reserve. For ease of exposition, we have chosen one to illustrate our detailed discussion of optimal sizing. All combinations of EDFs, market settings, and FD mandates are available from the authors.

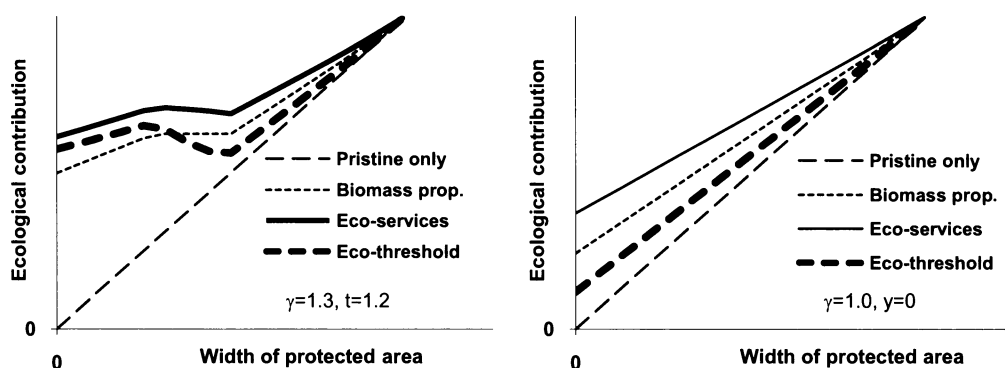


FIGURE 2

Ecological Contribution for Four Ecological Damage Functions under Two Market Settings as a Function of Protected Area Width

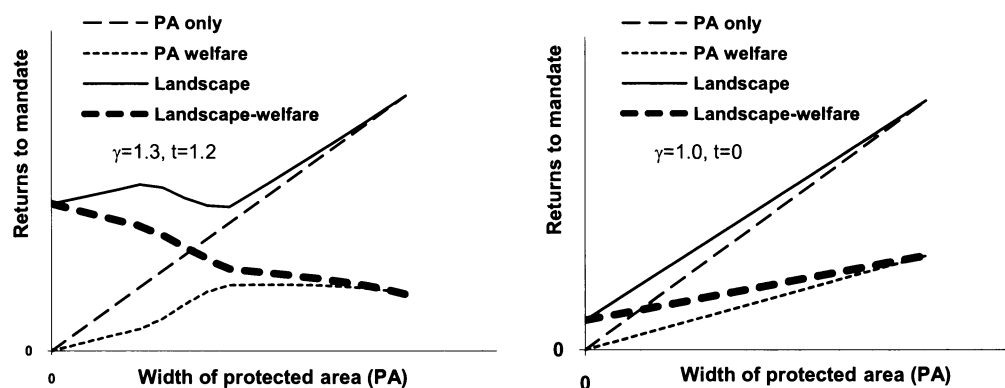


FIGURE 3

Returns to Four Forest Department Mandates under Two Market Settings for an Ecothreshold Ecological Damage Function

Because FDs are often constrained by limited budgets, and because of the highly nonlinear relationship between the width of the PA and the returns to a particular FD mandate, we show in Figure 3 returns to each FD mandate as a function of width of the PA, where the linear relationship between acquisition/enforcement costs and the width of the PA implies that the budget constraint translates to a PA width constraint.

When markets are well functioning, all FD mandates create the largest PA the budget allows, implying that with an unlimited enforcement budget the whole forest area is protected (Figure 3, right panel). However, when markets function less well (Figure 3, left panel),

whereas the FDs with a PA-only or landscape mandate will always protect the entire area, budget permitting, the FD with a PA-welfare mandate protects part of the forest and allocates part for extraction, and the FD with a landscape-welfare mandate gets sufficient ecological contribution from the unprotected zone combined with the impact on villagers that no PA is created but rather the whole area becomes an extractive reserve.

Figure 3 suggests that the local Amani manager's decision to allow a de facto buffer zone was appropriate, given his stated concerns over the impact of the PA on the villagers' welfare. However, if the FD's mandate were to change, such as in response to a rec-

ognition of the contribution of the unprotected forest to specific ecosystem services, the FD might well allow extraction from the full area of the reserve rather than from none (Figure 3, left panel). Conversely, if efforts were made to improve labor and NTFP markets, both the FD and the local Amani manager, whether concerned with only the PA or the full forested area, should be more likely to agree to protect the full area from extraction. By improving labor opportunities and market access, the marginal penalty imposed on the villagers is lowered, and such improvements typically, in our framework, increase the optimal size of the PA for FDs with a welfare mandate. Because of these interactions between the ecological and market settings and the mandate in determining the optimal size of the PA, policies addressing one component of the setting, such as rural development, can have a large impact on the optimal PA sizing decision, which implies that more integration of rural development and environmental policy could produce better welfare and ecological outcomes. For example, roads are often identified as a major driver of deforestation, yet in some scenarios depicted here, roads that provide access to markets might facilitate the establishment of larger PAs. Although roads may affect unprotected forests differently, without a framework for examining the interactions of the socioeconomic and ecological setting in determining PA size, the possibility of a positive conservation outcome from establishing roads is obscured.

In some situations, the use of a nonlandscape mandate may result from a lack of information about the value of degraded forests, that is, a lack of information about the EDF. In other cases, PA siting and sizing decisions may be made considering the landscape but without appropriate information about the EDF. Particularly when markets are poorly functioning, misunderstanding the EDF could result in suboptimal sizing decisions and unnecessary exclusion of villagers from forest reserves. For example, for our model parameterization, a landscape-welfare mandate leads to as large a PA as possible if the EDF is believed to be pristine-only, implying a high implicit penalty on villagers, but an extractive reserve if the EDF is believed to be ecotresh-

hold, which would impose no penalty on villagers. When FDs are operating under restricted budgets, which is often the case, miscalculating the EDF could cause further inefficiencies. We can see from Figure 3 that the returns to the landscape mandate include local maxima and minima that imply that at some budgets the optimal PA size can be smaller than the size that the budget allows. For example, a FD able to protect 60% of the width of the forest would choose to protect just under 10% of the forest if it thought the EDF to be biomass proportional or ecoservices, approximately 25% if it thought the EDF to be ecotreshold, but the full 60% if it thought the EDF to be pristine-only. Although identifying the optimal sizing decision under uncertainty about the EDF is out of the scope of this paper, the range of optimal PA sizes across different EDFs, market settings, budget levels, and mandates demonstrates the need for managers to assess the value of investing in information about the EDF and the setting before determining the PA size.

#### *PA Sizing Decisions and Ecological Information in Several Countries*

Our fieldwork observations suggest there has been less attention to the EDF in local forest management decisions than to the welfare considerations in Amani. But, if Tanzania follows the increasing emphasis on employing a landscape approach toward ecological conservation and rural development, such information about the EDF will become critical for the PA sizing decision.<sup>13</sup> Taking into account the landscape and the EDF might change the size of the local forest manager's extraction, or de facto buffer, zone. For example, if Amani is an ecotreshold EDF, a larger unprotected zone would improve welfare and ecological contributions, but if Amani is a fragile ecosystem where only pristine forest provides ecological benefits (pristine-only EDF), a smaller extraction zone would be

<sup>13</sup> Robinson and Albers (2006) recognize that, in reality, the extent to which PAs are in practice protected from all illegal activity also influences the optimal sizing decision. Albers (2010) demonstrates the interaction of buffer zone size, enforcement budget, and resource extraction production functions.

needed and the local forest manager might be better off putting more effort into tree planting activities rather than allowing continued extraction from the reserve.<sup>14</sup> Despite the reserve's importance, little is known about the EDF of the Amani forest—or many forests in developing countries—with respect to typical levels of extraction. Without specific information, from a landscape-welfare perspective (a reasonable proxy for the social optimum) whether the *de facto* buffer zone should be larger or smaller than that introduced in an *ad hoc* way by the local forest manager is not known. In general, in Tanzania the national perspective on PA size appears to center on ecological considerations, although in the absence of specific ecological information, while the local, implementation-level perspective in Amani appears to center on rural welfare and the PA-resources. Taking a landscape perspective and developing more ecological information will be particularly important to Tanzania as carbon storage and REDD policies become operational.

In Thailand, KYNP managers report that although tacitly permitted extraction does not appear to cause major damage to most ecosystem services of the unofficial buffer zone, extraction of some species, such as certain tree barks for making incense, threatens that species and its ecosystem functions. The EDF for an area may, therefore, not be identical for all of the services the ecosystem provides. If an organization attempts to size a PA without information about which EDF reflects that

ecological setting or the ecosystem services of concern, it will not make the correct sizing decision. That decision is most likely to be far from optimal when the true EDF reflects a threshold near the chosen size of the park, because both the ecological and the welfare losses of a too large or too small PA can be large in that setting. If KYNP managers view the ecosystem service of maintaining healthy incense-producing trees as critical, they might require more information about the EDF and about bark harvests to better protect that resource. In the case of Xishuangbanna Nature Reserve in China, managers permit extraction and herb planting for home use but appear to manage the PA with some recognition of the landscape and the relevant EDF, because they discuss the need for a high enough level of forest cover in the region to control humidity to support neighboring rubber plantations (Albers and Grinspoon 1997). In Kibaha's forests (including the Ruvu North and Ruvu South Forest Reserves) in Tanzania, the FD's actions also reflect a landscape-welfare mandate and a recognition of the relevant EDF in the area. The forest managers are well aware that ready access to urban markets has encouraged widespread extraction by local and nonlocal people for the sale of charcoal to nearby Dar es Salaam.<sup>15</sup> Regulations permit some extraction in the forest, which implies that the EDF is not perceived by policy makers to be pristine-only. Managers have responded to perceived overextraction by subsidizing and giving limited ownership rights to local people to plant trees in highly degraded areas in and outside of the reserve. These management decisions effectively reduce the size of the PA while protecting the ecological contribution of both the protected and unprotected forest areas. These decisions, and an albeit failed attempt by the Amani forest manager to encourage private tree planting, also indicate an added dimension in the issues surrounding "replacement" of extractive goods with purchases from the market versus displacement of extraction into non-PA forests by highlighting the importance of tree planting as a third op-

<sup>14</sup> Indeed, the *de facto* buffer zone in Amani was supposed to be temporary, lasting eight years, giving villagers time to plant trees on their own land, suggesting that the local forest manager was influenced either by the national regulation or an understanding of the EDF, but also by the high trade-offs between ecological benefits and villager welfare implied by a pristine-only EDF. However, after eight years, the buffer zone remained because of a lack of private tree planting but unchanging rural resource needs (forest officer, Amani Nature Reserve, 2008, personal communication). The decision to introduce an eight-year temporary buffer zone recognized implicit dynamics of sizing and siting decisions that are beyond the scope of this paper. Conceptually the buffer zone was to mitigate the costs imposed on the villagers through the reintroduction of the PA while the villagers grew trees to substitute for the loss of access. However, dynamic inconsistency also played its part: after eight years, villagers were still predominantly relying on the buffer zone for NTFPs, and so the buffer zone was renewed.

<sup>15</sup> Lema Mathias, project manager, Ruvu-Fuelwood Project, Kibaha, 2008, personal communication.

tion, especially when purchasing from the market may simply imply an indirect displacement from more distant forests.

#### IV. DISCUSSION AND CONCLUSION

Externalities resulting from the introduction or reintroduction of a PA, in the form of what are variously described as NTFP extraction spillovers, leakages, or displacement, have long been recognized in the literature. Yet information about how villagers displace their extraction to other areas, paired with an understanding of the EDF, are not often used to inform PA sizing decisions.

The model we develop in this paper does not yet encompass all aspects of PA management. A next research step might combine our villager and socioeconomic-based perspective with other functional forms suggested by the ecology literature for the relationship between the ecological benefits and PA size (Diamond 1975). Because incomplete enforcement is a further source of leakage from PAs, combining the villager-FD model here with the models presented by Robinson and Albers (2006), Robinson and Lokina (2011), and Albers (2010) that focus on incomplete enforcement and pressure on PA boundaries would inform the joint decision of sizing and enforcement. A similar extension would involve making the level of access restrictions a choice variable in addition to size, leading to results about where to locate more- and less-restrictive parks with respect to market access. Ours is a static model, sufficient to demonstrate the key points of this paper, but a spatial-temporal model of extraction and resource regeneration, as given by Robinson and Lokina (2011) and Robinson, Albers, and Williams (2008), could lead to advice about varying the optimal size of a PA over time or about varying the optimal enforcement of that PA over time.

Our model, by abstracting from some of these aspects of PA management, highlights complex interactions of ecological and socioeconomic characteristics. First, the displacement of extraction into buffer zones and unprotected forests following the establishment of a PA creates a classic spatial spillover, which we identify as a “displacement” effect. Because a villager’s response to a PA alters

the quality of the resource in the area just outside of the reserve, that area is not an appropriate “control” for evaluating the success of a reserve. Such criticisms have been leveled at various studies that perform inside PA–outside PA deforestation comparisons, such as Bruner et al.’s (2001) use of 10-km-wide bands adjoining parks. Halpern, Gaines, and Warner (2004, 1253) find something similar for marine reserves and state that “no particular area outside a reserve can be reliably identified as a control until we have a much better understanding of the dispersal distances of pelagic larvae.” Similar statements can be made about human extractors. Our paper suggests that the spillovers could be large if markets function poorly, or small if markets function well. In the former case the area just outside the PA would be a particularly poor control.

Second, complementing our focus on spatial spillovers in forest degradation, economists are beginning to address the spatial spillovers in deforestation that result from PAs. Recent econometric approaches using propensity score matching have attempted to find appropriate controls outside of parks to evaluate how effective parks are in preventing deforestation. Andam et al. (2008) find in Costa Rica that not only are protected forests effective in reducing deforestation, but spillovers from protected to unprotected forests are negligible. Such a finding fits with our model’s prediction that where markets are relatively well developed, as in Costa Rica as compared to Tanzania, spillovers and welfare effects are small. Pfaff et al. (2007) find that the deforestation spillovers from parks in Brazil depend on proximity to major urban centers. Our framework tacitly assumes equal population densities in near-market and distant-market settings, and so the comparison to Pfaff et al.’s work is not exact, but their work underscores the need for understanding the socioeconomic setting in predicting the magnitude of spillovers. Viewed through the lens of our framework, these empirical papers suggest that more modeling work into the spatial/location aspect of the deforestation decision is needed, as is a distinction among deforestation for agriculture, timber, ranching, or urbanization, if the landscape as a whole



matters. Because deforestation dramatically reduces the ecological services of the land, the particular EDF may prove less important than the market and socioeconomic setting in which deforestation decisions are made.

Third, our focus on villagers' responses to a PA in terms of where and what quantities of NTFPs they collect, the relationship between this response and the ecological contribution of the forested landscape, and the size of spatial spillovers as a function of markets has particular relevance to the REDD discussion's emphasis on the significant contribution of forest degradation to greenhouse gas emissions. The REDD literature recognizes that degradation occurs according to a different process from deforestation (Murdiyarso et al. 2008). Widespread monitoring or detection of degradation poses difficulties with currently available technologies, which raises the idea of using probabilistic models to characterize the risk of degradation in different locations. Our paper provides a behavioral model interacting with characteristics of the ecological and socioeconomic setting that could form the foundation for such probabilistic models of degradation risk. Simpler frameworks might miss the interactions among EDFs, market access, labor opportunities, and conservation efforts.

Fourth, our paper makes explicit the importance of understanding markets. For example, for the same-sized forest, policy makers can create larger exclusion areas in forests close to markets, as compared to more remote areas where people typically use and degrade forests outside of reserves to meet consumption needs. Because our model does not account for rural population densities and differences in enforcement costs and budgets across regions, our prediction reflects only a piece of the story, but it does draw attention to the role of market access.<sup>16</sup> Policy makers

do not regularly incorporate market information into their PA sizing decisions, perhaps in part because few researchers provide that information. Otherwise carefully conducted surveys within villages near PAs (e.g., Rao et al. 2003; Allendorf et al. 2006; Shrestha and Alavalapati 2006; Spiteri and Nepal 2008) do not include questions about the local markets for NTFPs. Similarly, careful studies of local markets for NTFPs, such as by Ghate, Mehra, and Nagendra (2009), do not inquire about local labor markets.

Fifth, forest managers cannot make appropriate landscape-level conservation plans without information about the value of environmental services provided by forests at varying levels of degradation. Our analysis is a call for more research to characterize the provision of environmental services as a function of the forest's degradation. For example, the optimal proportion of the forest that is protected can flip from zero to 100%, depending on the applicable EDF, especially if the EDF contains a threshold. Although these findings may appear intuitive, ecologists and valuation economists rarely provide the information necessary to distinguish among these possibilities. In the absence of information about the relevant EDF in a particular setting, policy makers could perform a sensitivity analysis over a range of EDF heights and shapes.

Finally, our paper shows that more than species and ecology should enter the decision about optimal PA size, because rural people in poor countries react to the establishment and enforcement of a PA. That reaction interacts with the ecological and socioeconomic setting to determine the optimal PA size for a given mandate. We have observed in Tanzania and elsewhere that local forest managers often alter the size of their PAs. That observation, paired with our model, suggests that national-level PA mandates should be informed by local considerations, rather than being determined by ecological conditions in isolation from local socioeconomic conditions. While most of the reserve literature focuses on the ecological aspects or on the social aspects of park siting, sizing, and management questions, our analysis demonstrates that the optimal PA size does not correspond to a simple averaging or summing up of the individual

<sup>16</sup> In practice, population density may be higher near markets than in more remote areas. This model says that areas without market access will see larger amounts of extraction/degradation per individual than areas with market access. In market access areas, then, one might expect to see more people extracting but each person extracting less than in no-market areas, creating a trade-off for policy makers to consider in siting PAs. Pfaff et al. (2009) report on such relationships for Costa Rica.

components: Villager reactions are a function of the market setting; both welfare and ecological outcomes are functions of the ecological setting and the villager reaction; and the market setting, the ecological setting, and the FD mandate interact in nonlinear ways.

## APPENDIX 1: VILLAGER REACTION FUNCTIONS

Following Robinson, Williams, and Albers (2002), our representative villager maximizes returns to her labor, choosing how much time to spend collecting NTFPs from the unprotected zone, and how much time on other activities, subject to an NTFP consumption requirement  $R$ . To represent an important subset of resource extraction, villagers must meet a consumption requirement. Fuelwood is an example of an extracted resource for which households in practice have something close to a minimum need. Pattanayak, Sills, and Kramer (2004) found that fuelwood is an essential good whose demand is highly inelastic. Gosalamang, Pal, Gombya-Ssembajjwe (2004) found that in Uganda, villagers who were excluded from the Mount Elgon Forest Reserve continued to extract fuelwood for subsistence purposes but stopped collecting fuelwood for sale and greatly reduced their extraction of nonessentials. This appendix focuses on how forest resources and the value of the forest are affected when a villager is constrained by the PA, such that the villager extracts throughout the full width of the unprotected zone to the edge of the PA. The villager chooses to what extent she meets her consumption requirement from the unprotected zone and to what extent she relies on the NTFP market. Her key choice variable is  $w$ , the time spent extracting per unit distance  $w$ —the greater  $w$ , the greater the harvest per unit distance. This variable translates into the intensity of harvest extraction throughout the unprotected zone,  $h(w, m, \alpha)$ , where  $m$  is the initial resource density and  $\alpha$  is a parameter that takes into account the effectiveness of the villager's extraction effort. Because  $m$  is constant across distance, as derived by Robinson, Williams, and Albers (2002), so too are  $w$  and  $h$ . Distance is a cost to the villager, and so we also have to take explicit account of the time it takes the villager to walk a unit distance through the unprotected zone when not extracting,  $v$ . The total harvest  $H$  is  $hX_B$ , and the total time spent in the forest is  $(w + v)X_B$  at a time cost  $C[(w + v)X_B]$ . Our model is in keeping with other household resource extraction models in its use of scarce (and costly) labor time as the primary input to producing the extracted forest product (Pattanayak, Sills, and Kramer 2004; Kohlin and Parks 2001; Amacher, Hyde, and Kanel 1996;

Bluffstone 1995). Any extracted surplus  $S$  is sold to the market, and any deficit  $D$  is purchased, at a price  $p$ , but in each case, transportation costs  $t$  are incurred (Omamo 1998; Key, Sadoulet, and de Janvry 2000). The parameter  $t$  can also proxy for the difficulty of interacting with a cash-based market with villagers who largely participate in subsistence activities (On-gugo, Njuguna, and Riako 2004). In sum, the villager contemplates the following single-period optimization of net returns  $V$ , which translates directly into the villager's welfare:

$$\begin{aligned} \max_w \{V\} &= \max_w \{(p - t)S - (p + t)D - C\} \\ \text{s.t. } D - S &= R - H; \quad H, S, D \geq 0; \quad SD = 0; \\ H &= hX_B, \quad C = C[(w + v)X_B]. \end{aligned} \quad [A1]$$

To solve the model explicitly, we chose functional forms for the harvest and cost functions that are simple enough to permit analytical solutions while maintaining the required characteristics in terms of first and second-order derivatives:

$$h = m \left( 1 - \frac{1}{(1 + \alpha mw)} \right) \quad \text{and} \quad C = k[(w + v)X_B]^\gamma, \quad [A2]$$

where  $k$  is simply a scale factor and  $\gamma \geq 1$ , implying a constant or increasing marginal opportunity cost of time.

A villager's endogenous type when the exclusion zone is in place may be "subsistence," extracting exactly the requirement  $R$ ; "supplementing," purchasing some of her requirement from the market; or "selling," selling an excess over and above the requirement to the market. A fourth type, "nonextractor," is a subset of supplementing villagers, where a villager purchases her full requirement from the market.

If the representative villager when constrained by the exclusion zone, or the PA, is a subsistence villager,  $S = D = 0$ :

$$H = R = h(\bar{X} - X_P) = m \left[ 1 - \frac{1}{(1 + \alpha mw)} \right] (\bar{X} - X_P),$$

and so,

$$h = \frac{R}{\bar{X} - X_P} \quad \text{and} \quad w = \frac{R}{\alpha m [\bar{X} - X_P - R]}. \quad [A3]$$

If the representative villager when constrained is a supplementing villager,  $S = 0$  and

$$D = R - H = R - h(\bar{X} - X_p).$$

$w$  is the solution to the first-order condition:

$$(p + t)\alpha m^2 = k\gamma[(w + v)(\bar{X} - X_p)]^{\gamma-1}(1 + \alpha mw)^2. \quad [A4]$$

Similarly, if she is a selling villager,  $D = 0$  and  $S = H - R = h(\bar{X} - X_p) - R$ :

$$(p - t)\alpha m^2 = k\gamma[(w + v)(\bar{X} - X_p)]^{\gamma-1}(1 + \alpha mw)^2. \quad [A5]$$

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